26900067 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCBOT301 - PLANT CELL AND MOLECULAR BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **1** marks

1. Name the scientists who proposed cell theory. [K] / [CO1]
2. Define telomere. [U] / [CO3]
3. A cell shrinks and breaks into membrane-bound fragments without affecting the neighbouring cells. Identify the process. [An] / [CO5]
4. State the function of RNA polymerase. [U] / [CO5]
5. Define RNA splicing. [K] / [CO5]
6. Identify which phase of the cell cycle shows the greatest overall cell growth before DNA synthesis begins. [An] / [CO5]
7. What is the functional role of DNA ligases in DNA replication? [K] / [CO6]
8. What type of bond holds the two strands of DNA together? [U] / [CO5]

9. Describe the basic structure of the nuclear pore complex. [U] / [CO3]

10. What is the process by which an ancestral cell engulfed a bacterium called? [K] / [CO6]

11. Who formulated the base-pairing rules for DNA? [U] / [CO5]

12. What process allows lysosomes to degrade worn-out intracellular organelles? [K] / [CO3]

[1x10 = 10]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain the significance of conserved genes and proteins citing suitable examples. [U] / [CO5]

14. Illustrate and explain the salivary gland chromosome. [U] / [CO3]

15. Describe the wobble hypothesis proposed by Crick. [U] / [CO4]

16. Write a comparative note on heterochromatin and euchromatin. [K] / [CO3]

17. Discuss how the cytoskeleton helps in cell structure and function. [U] / [CO2]

18. What are frame-shift mutations? Explain the significance of frame-shift mutations. [K] / [CO6]

19. DNA repair mechanisms in cells are essential for the normal well-being of an organism. Why? [U] / [CO6]

20. Describe the structure and functions of mitochondria in a plant cell. [U] / [CO2]

[5x6 = 30]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

21. Describe the basic chemical composition of the plant cell membrane. [U] / [CO2]

22. Write an essay on different types of point mutations. [K] / [CO6]

[10x1 = 10]